

Supplementary Information (SI)

**Rational Design and Modular Synthesis of Biodegradable Ionizable Lipids via Passerini
Reaction for mRNA delivery**

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Section A. Supplementary Methods

Synthesis and characterization of combinatorial ionizable lipid library.

Synthesis of 1-isocyanoundecane (A1):

1-isocyanoundecane was synthesized by the same procedure as our previous work described^{1,2}. In a 250 mL round-bottomed flask, a mixture of 82.75 mL of tetrahydrofuran (THF) and 2.2511 g of solid sodium formate was stirred to create a suspension. Concurrently, a solution of acetyl chloride (1.0 M) in dichloromethane (34.755 mL) was slowly added dropwise to the above-prepared suspension. The resulting mixture was stirred at room temperature for 12 hours, yielding reaction solution 1. Subsequently, compound undecan-1-amine was dissolved in tetrahydrofuran (THF) and added dropwise to reaction solution 1, followed by stirring at room temperature for an additional 4 hours to obtain reaction solution 2. Reaction solution 2 was then diluted with 125 mL of deionized water, and the aqueous phase was extracted twice with 240 mL of ethyl acetate (120 mL each time). The resulting extract underwent drying with anhydrous magnesium sulfate (MgSO₄) for 0.5 hours, followed by spin-drying to yield the crude *N*-undecylformamide. To the crude *N*-undecylformamide product, a solution of THF (50 mL) was added to a 100 mL round-bottomed flask, and the mixture was placed in an ice bath at 0°C. After stirring for 10 minutes, triethylamine (TEA) was introduced, and the mixture was stirred for an additional 15 minutes in the ice bath to obtain reaction solution 1. In a separate preparation, solution 1 was formed by combining THF (12.9 mL) and phosphorus oxychloride (3.519 g). Under ice bath conditions, solution 1 was gradually added dropwise to reaction solution 1 using a constant pressure-dropping funnel. The resulting reaction mixture was stirred for 1.5 hours. To dilute and wash the mixture, 65 mL of saturated sodium bicarbonate solution was added, and the washed solution was extracted twice with 130 mL of ether (65 mL each time). The combined extracts were treated with anhydrous MgSO₄, stirred, and dried for 0.5 hours. The solvent was then removed by rotary evaporation, and the crude product was purified using a column with a Hexane/ethyl acetate system to afford the compound A1 as a yellowish oil (Yield 83%). ¹H NMR (400 MHz, CDCl₃) δ 3.42 – 3.27 (m, 2H), 1.63 – 1.52 (m, 2H), 1.31 – 1.24 (m, 16H), 0.88 – 0.86 (m, 3H). MS (*m/z*): MW calc'd for C₁₂H₂₃N (M + H⁺): 181.18, found: 181.68.

Synthesis of 1-isocyanododecane (A2):

Same method as A1 to afford the compound A2 as a yellowish oil (Yield 75%). ¹H NMR (400 MHz, CDCl₃) δ 3.49 – 3.22 (m, 2H), 1.72 – 1.57 (m, 2H), 1.50 – 1.34 (m, 2H), 1.26 (d, *J* = 10.0 Hz, 16H), 0.89 – 0.84 (m, 3H). MS (*m/z*): MW calc'd for C₁₃H₂₅N (M + H⁺): 195.20, found: 195.35.

Synthesis of 1-isocyanohexadecane (A3):

Same method as A1 to afford the compound A3 as a yellowish oil (Yield 78%). ¹H NMR (400 MHz, CDCl₃) δ 3.45 – 3.26 (m, 2H), 1.66 (ddtt, *J* = 11.4, 6.7, 4.6, 2.3 Hz, 2H), 1.46 – 1.38 (m, 2H), 1.26 (d, *J* = 11.1 Hz, 24H), 0.87 (t, *J* = 6.8 Hz, 3H). MS (*m/z*): MW calc'd for C₁₇H₃₃N (M + H⁺): 251.26, found: 251.67.

Synthesis of Tail (Z)-1-isocyanooctadec-9-ene (A4):

Same method as A1 to afford the compound A4 as a yellowish oil (Yield 72%). ¹H NMR (400 MHz, CDCl₃) δ 5.41 – 5.21 (m, 2H), 3.45 – 3.29 (m, 2H), 1.99 (dq, *J* = 14.2, 6.0 Hz, 3H), 1.66 (ddd, *J* = 8.4, 5.6, 3.4 Hz, 2H), 1.48 – 1.35 (m, 2H), 1.28 (dt, *J* = 17.3, 6.2 Hz, 20H), 0.92 – 0.84 (m, 3H). MS (*m/z*): MW calc'd for C₁₉H₃₅N (M + H⁺): 277.28, found: 277.47.

Synthesis of decanal (B1):

To create the decanal, dissolve the decanol in DCM and add 1.2 times the amount of Dess–Martin periodinane (DMP) for 2h. After the solvent was removed by rotary evaporation under a vacuum, it was purified through the column (Hexane/EA = 80: 20) to afford the compound B1 as a colorless oil (Yield 85%). ¹H NMR (400 MHz, CDCl₃) δ 9.76 (t, *J* = 1.9 Hz, 1H), 2.42 (td, *J* = 7.4, 1.9 Hz, 2H), 1.62 (p, *J* = 7.2 Hz, 2H), 1.41 – 1.14 (m, 12H), 0.99 – 0.68 (m, 3H). MS (*m/z*): MW calc'd for C₁₀H₂₀O (M + H⁺): 156.15, found: 156.22.

Synthesis of olealdehyde (B2):

Same method as B1 to afford the compound B2 as a colorless oil (Yield 82%). ¹H NMR (400 MHz, CDCl₃) δ 9.75 (t, *J* = 1.9 Hz, 1H), δ 5.51 – 5.20 (m, 2H), 2.39 (dt, *J* = 36.8, 7.5 Hz, 2H), 2.19 – 1.86 (m, 4H), 1.62 (q, *J* = 7.5 Hz, 2H), 1.42 – 1.14 (m, 20H), 0.91 – 0.83 (m, 3H). MS (*m/z*): MW calc'd for C₁₈H₃₄O (M + H⁺): 266.26, found: 266.42.

Synthesis of 6-oxohexyl dodecanoate (B3):

Dissolve dodecanoic acid (2.0 g, 7.80mmol), hexane-1,6-diol (1.84 g, 15.60mmol), N, N'-Dicyclohexylcarbodiimide (DCC) (2.41 g, 11.70 mmol), and 4-(dimethylamino) pyridine (DMAP) (95.29 mg, 779.94μmol) in order in a 100 mL round-bottomed flask containing 50 mL DCM and mix well at room temperature for 24 hours. To create the aldehyde,

dissolve the intermediate in DCM and add 1.2 times the amount of DMP for 2h. After the solvent was removed by rotary evaporation under a vacuum, it was purified through the column (Hexane/EA= 80: 20) to afford the compound B3 as a colorless oil (Yield 78%). ¹H NMR (400 MHz, CDCl₃) δ 9.76 (t, *J* = 1.7 Hz, 1H), 4.06 (t, *J* = 6.6 Hz, 2H), 2.45 (td, *J* = 7.3, 1.7 Hz, 2H), 2.28 (t, *J* = 7.6 Hz, 2H), 1.75 – 1.54 (m, 6H), 1.44 – 1.36 (m, 2H), 1.30 – 1.21 (m, 16H), 0.88 (dd, *J* = 6.7, 1.7 Hz, 3H). MS (*m/z*): MW calc'd for C₁₈H₃₄O₃ (M + H⁺): 298.25, found: 298.33.

Synthesis of 6-oxohexyl 2-hexyl decanoate (B4):

Same method as B3 to afford the compound B4 as a colorless oil (Yield 67%). ¹H NMR (400 MHz, CDCl₃) δ 9.76 (t, *J* = 1.7 Hz, 1H), 4.06 (t, *J* = 6.6 Hz, 2H), 2.44 (td, *J* = 7.3, 1.6 Hz, 2H), 2.30 (tt, *J* = 9.0, 5.3 Hz, 1H), 1.76 – 1.54 (m, 6H), 1.42 – 1.21 (m, 23H), 0.88 – 0.83 (m, 6H). MS (*m/z*): MW calc'd for C₂₂H₄₂O₃ (M + H⁺): 354.31, found: 354.58.

Synthesis of 6-oxohexyl 2-butyloctanoate:

Same method as B3 to afford the compound 6-oxohexyl 2-butyloctanoate as a colorless oil (Yield 55%). ¹H NMR (400 MHz, CDCl₃) δ 4.06 (t, *J* = 6.6 Hz, 2H), 2.53 – 2.38 (m, 2H), 2.29 (tt, *J* = 9.0, 5.4 Hz, 1H), 1.74 – 1.50 (m, 6H), 1.48 – 1.35 (m, 4H), 1.33 – 1.17 (m, 13H), 0.86 (td, *J* = 7.0, 2.2 Hz, 6H). MS (*m/z*): MW calc'd for C₁₈H₃₅O₃ (M + H⁺): 299.47, found: 299.27.

Synthesis of Lipid of (Z)-7-((3-(dimethylamino)propanoyl)oxy)-8-(octadec-9-en-1-ylamino)-8-oxooctyl dodecanoate:

The synthesis of **H2A4B3** is the same as above. A mixture of 3-(dimethylamino)propanoic acid (30 mg, 256.09 μmol), (Z)-1-isocyanooctadec-9-ene (71.06 mg, 256.09 μmol) and 6-oxohexyl dodecanoate (76.43 mg, 256.09 μmol) in anhydrous solvent DCM (1 mL) and was stirred in capped glass vials at room temperature for 12 h. Purification with a gradient (1% Ammonia, 10% MeOH, 89% DCM) and getting the final product as a colourless oil (Yield: 150 mg, 84%). R_f = 0.4 (Ammonia/MeOH/DCM=1:10:89). ¹H NMR (400 MHz, CDCl₃) δ 5.51 – 5.11 (m, 3H), 4.08 – 3.94 (m, 2H), 3.34 – 3.10 (m, 2H), 2.85 – 2.10 (m, 6H), 2.05 – 1.76 (m, 6H), 1.61 (dp, *J* = 23.9, 7.8 Hz, 7H), 1.25 (d, *J* = 6.6 Hz, 50H), 0.88 – 0.85 (m, 6H). MS (*m/z*): [M+H]⁺ calc'd for C₄₂H₈₁N₂O₅⁺ 694.11, found: 694.80.

Synthesis of Lipid of (Z)-7-((3-(dimethylamino)propanoyl)oxy)-8-(octadec-9-en-1-ylamino)-8-oxooctyl 2-hexyldecanoate:

The synthesis of **H2A4B4** through 3CR-Passerini reaction. A mixture of 3-(dimethylamine) propanoic acid (30 mg, 256.09 μmol), (Z)-1-isocyanooctadec-9-ene (71.06 mg, 256.09 μmol) and 6-oxohexyl 2-hexyl decanoate (90.8 mg, 256.09 μmol) in anhydrous solvent DCM (1 mL) and was stirred in capped glass vials at room temperature for 12 h. Further purification is achieved by flash column chromatography on a Combiflash system eluting with a gradient (1% Ammonia, 10% MeOH, 89% DCM) and getting the final product as a colourless oil (Yield: 152 mg, 78%). R_f = 0.4 (Ammonia/MeOH/DCM=1:10:89). ¹H NMR (400 MHz, CDCl₃) δ 5.32 (d, *J* = 16.3 Hz, 2H), 5.20 (dd, *J* = 7.9, 3.7 Hz, 1H), 4.07 – 4.04 (m, 2H), 3.72 – 3.45 (m, 2H), 3.21 (dd, *J* = 45.0, 7.5 Hz, 2H), 2.56 – 2.37 (m, 2H), 2.34 – 2.20 (m, 8H), 2.07 – 1.88 (m, 5H), 1.25 (m, 49H), 0.92 – 0.83 (m, 9H). MS (*m/z*): [M+H]⁺ calc'd for C₄₆H₈₈N₂O₅⁺ 750.22, found: 750.55.

Synthesis of Lipid of (Z)-1-(((Z)-octadec-9-en-1-yl)amino)-1-oxononadec-10-en-2-yl 3-(dimethylamino)propanoate:

The synthesis of **H2A4-Ole** through 3CR-Passerini reaction. A mixture of 3-(dimethylamine) propanoic acid (30 mg, 256.09 μmol), (Z)-1-isocyanooctadec-9-ene (71.06 mg, 256.09 μmol) and olealdehyde (68.24 mg, 256.09 μmol) in anhydrous solvent DCM (1 mL) and was stirred in capped glass vials at room temperature for 12 h. Further purification is achieved by flash column chromatography on a Combiflash system eluting with a gradient (1% Ammonia, 5% MeOH, 94% DCM) and getting the final product as a colourless oil (Yield: 132 mg, 82.28%). R_f = 0.3 (Ammonia/MeOH/DCM=1:5:94). ¹H NMR (400 MHz, CDCl₃) δ 5.34 (s, 5H), 3.33 – 3.25 (m, 2H), 2.00 (d, *J* = 6.5 Hz, 4H), 1.58 (s, 6H), 1.41 – 1.21 (m, 56H), 0.90 – 0.77 (m, 6H). MS (*m/z*): [M+H]⁺ calc'd for C₄₂H₈₁N₂O₃⁺ 662.11, found: 662.34.

Synthesis of Lipid of (Z)-6-((3-(dimethylamino)propanoyl)oxy)-7-(octadec-9-en-1-ylamino)-7-oxoheptyl 2-butyloctanoate:

The synthesis of **H2A4-Butyl** through 3CR-Passerini reaction. A mixture of 3-(dimethylamine) propanoic acid (30 mg, 256.09 μmol), (Z)-1-isocyanooctadec-9-ene (71.06 mg, 256.09 μmol) and 6-oxohexyl 2-butyl octanoate (76.43 mg, 256.09 μmol) in anhydrous solvent DCM (1 mL) and was stirred in capped glass vials at room temperature for 12 h. Further purification is achieved by flash column chromatography on a Combiflash system eluting with a gradient (1% Ammonia, 5% MeOH, 94% DCM) and getting the final product as a colourless oil (Yield: 147 mg, 82.28%). R_f

121 = 0.3 (Ammonia/MeOH/DCM=1:5:94). ¹H NMR (400 MHz, CDCl₃) δ 8.16 (d, *J* = 1.8 Hz, 1H), 5.35 (q, *J* = 3.1 Hz,
122 3H), 4.15 – 3.94 (m, 2H), 3.29 (d, *J* = 6.8 Hz, 2H), 2.01 (d, *J* = 6.3 Hz, 4H), 1.53 – 1.37 (m, 7H), 1.31 – 1.23 (m, 50H),
123 0.87 (d, *J* = 2.6 Hz, 6H). MS (m/z): [M+H]⁺ calc'd for C₄₂H₈₁N₂O₅⁺ 694.11, found: 694.15.

124 *Synthesis of Lipid of (Z)-6-((4-(dimethylamino)butanoyl)oxy)-7-(octadec-9-en-1-ylamino)-7-oxoheptyl 2-*
125 *hexyldecanoate:*

126 The synthesis of **A4B4-S3** is the same as above. A mixture of 4-(dimethylamino)butanoic acid (30 mg, 228.7 μmol),
127 (Z)-1-isocyanooctadec-9-ene (81.09 mg, 228.7 μmol) and 6-oxohexyl dodecanoate (63.64 mg, 228.7 μmol) in
128 anhydrous solvent DCM (1 mL) and was stirred in capped glass vials at room temperature for 12 h. Purification with
129 a gradient (1% Ammonia, 10% MeOH, 89% DCM) and getting the final product as a colourless oil (Yield: 148 mg,
130 85%). R_f = 0.4 (Ammonia/MeOH/DCM=1:10:89). ¹H NMR (400 MHz, CDCl₃) δ 5.38 – 5.33 (m, 3H), 4.07 (s, 2H),
131 3.50 – 3.45 (m, 2H), 2.77 (t, *J* = 6.6 Hz, 2H), 2.05 (t, *J* = 6.5 Hz, 3H), 1.93 (dd, *J* = 12.9, 3.5 Hz, 7H), 1.73 – 1.54 (m,
132 19H), 1.34 – 1.12 (m, 50H), 0.87 (dd, *J* = 7.3, 4.9 Hz, 9H). ¹³C NMR (126 MHz, cdcl₃) δ 176.95, 164.69, 161.27,
133 130.12, 129.89, 129.84, 72.06, 64.00, 45.96, 41.88, 39.27, 38.34, 34.93, 32.73, 32.70, 32.63, 32.05, 32.03, 32.01,
134 31.99, 31.91, 31.82, 31.38, 29.89, 29.87, 29.85, 29.83, 29.81, 29.80, 29.79, 29.77, 29.75, 29.73, 29.70, 29.69, 29.66,
135 29.65, 29.62, 29.59, 29.58, 29.55, 29.52, 29.50, 29.49, 29.47, 29.45, 29.40, 29.39, 29.37, 29.34, 29.33, 29.31, 29.30,
136 29.27, 29.25, 29.12, 28.74, 27.59, 27.55, 27.35, 27.31, 26.97, 26.52, 25.77, 24.67, 22.82, 22.81, 22.79, 22.72, 14.24,
137 14.23, 14.20. MS (m/z): [M+H]⁺ calc'd for C₄₇H₉₀N₂O₅⁺ 763.68, found: 763.49.

138 *Synthesis of Lipid of (Z)-6-((5-(dimethylamino)pentanoyl)oxy)-7-(octadec-9-en-1-ylamino)-7-oxoheptyl 2-*
139 *hexyldecanoate:*

140 The synthesis of **A4B4-S4** is the same as above. A mixture of 5-(dimethylamino)pentanoic acid (30 mg, 206.6 μmol),
141 (Z)-1-isocyanooctadec-9-ene (57.33 mg, 206.6 μmol) and 6-oxohexyl dodecanoate (73.26 mg, 206.6 μmol) in
142 anhydrous solvent DCM (1 mL) and was stirred in capped glass vials at room temperature for 12 h. Purification with
143 a gradient (1% Ammonia, 10% MeOH, 89% DCM) and getting the final product as a colorless oil (Yield: 136 mg,
144 84%). R_f = 0.4 (Ammonia/MeOH/DCM=1:10:89). ¹H NMR (400 MHz, CDCl₃) δ 8.22 (d, *J* = 1.7 Hz, 1H), 5.40 (q, *J*
145 = 7.4 Hz, 3H), 4.23 – 3.66 (m, 2H), 3.30 (dq, *J* = 34.7, 6.8 Hz, 4H), 2.47 – 1.78 (m, 9H), 1.73 – 1.23 (m, 65H), 1.02
146 – 0.72 (m, 9H). MS (m/z): [M+H]⁺ calc'd for C₄₈H₉₂N₂O₅⁺ 777.70, found: 777.82.

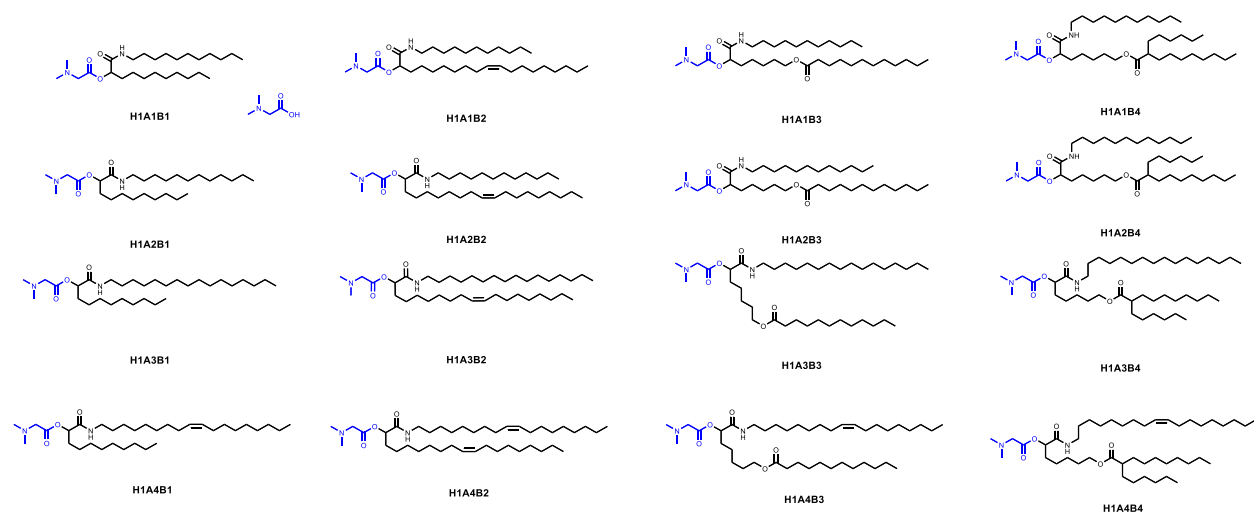


Figure S1. Lipid structures based on the head group dimethylglycine (H1) and synthesized by the Passerini reaction.

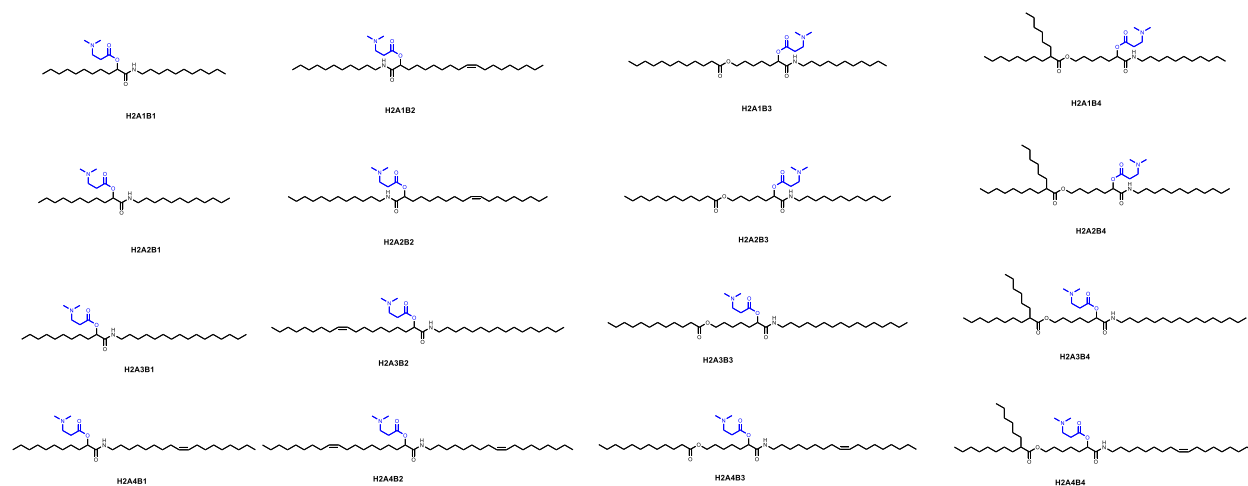


Figure S2. Lipid structures based on the head group 3-(dimethyl amino) propanoic acid (H2) and synthesized by the Passerini reaction.

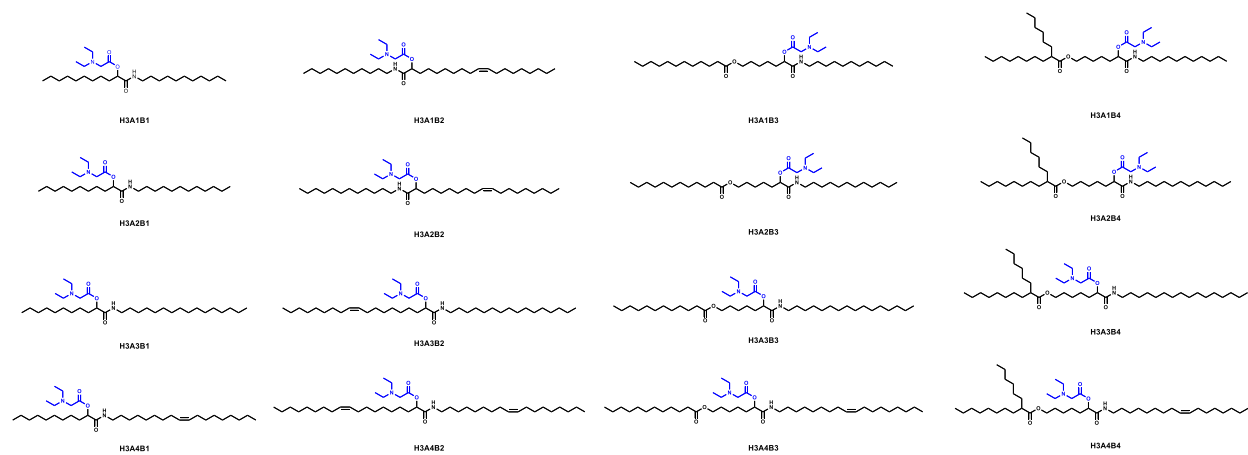


Figure S3. Lipid structures based on the head group dimethylglycine (H3) and synthesized by the Passerini reaction.

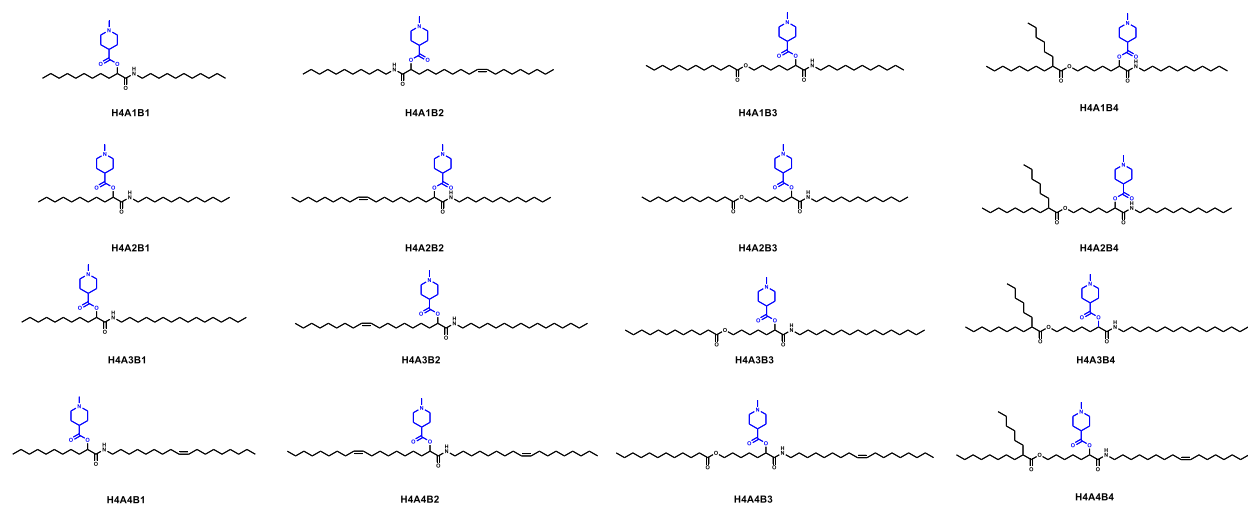


Figure S4. Lipid structures based on the head group 1-methylpiperidine-4-carboxylic acid (H4) and synthesized by the Passerini reaction.

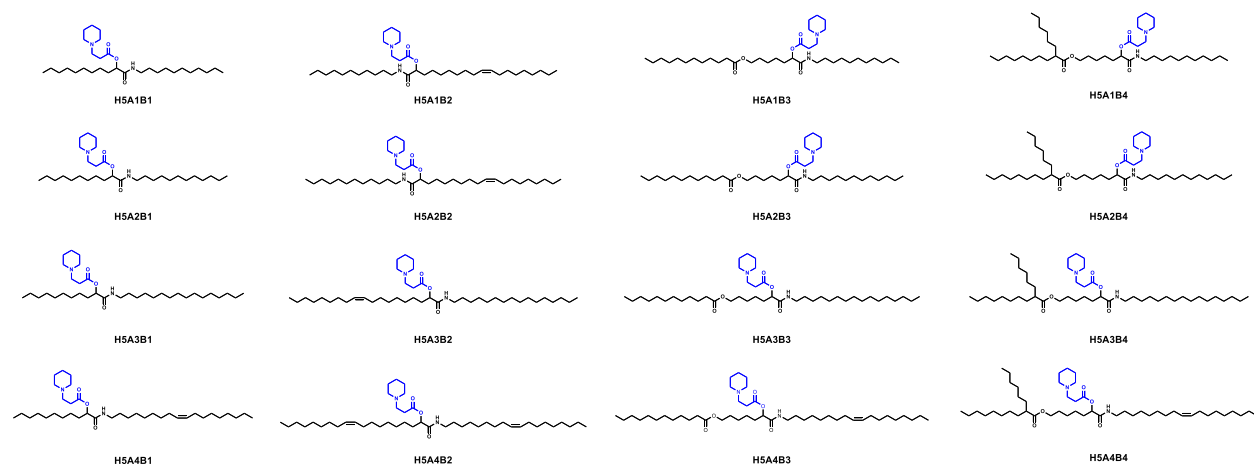


Figure S5. Lipid structures based on the head group 3-(piperidin-1-yl) propanoic acid (H5) and synthesized by the Passerini reaction.

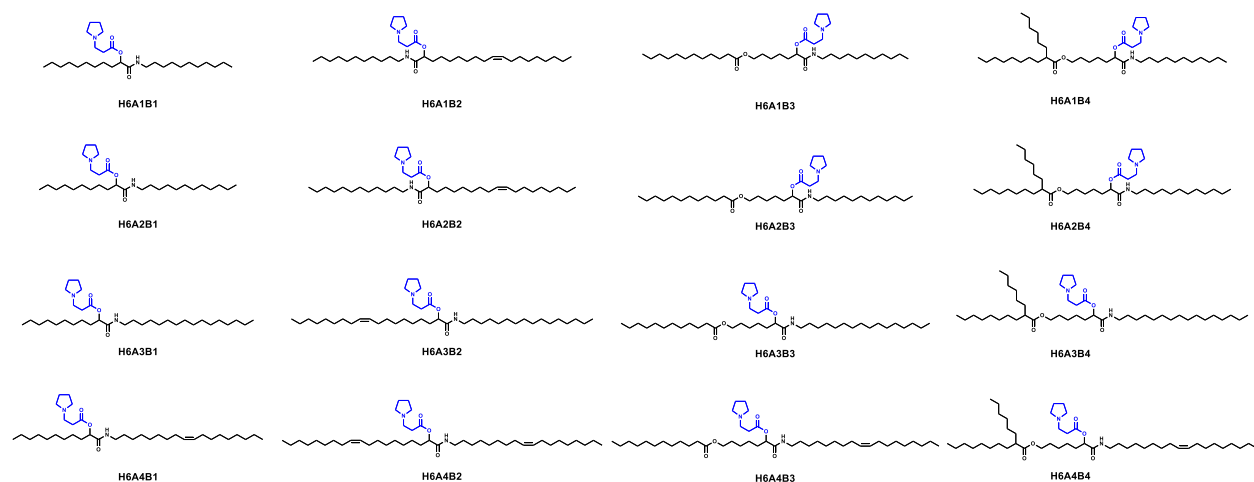


Figure S6. Lipid structures based on the head group 3-(pyrrolidin-1-yl) propanoic acid (H6) and synthesized by the Passerini reaction.

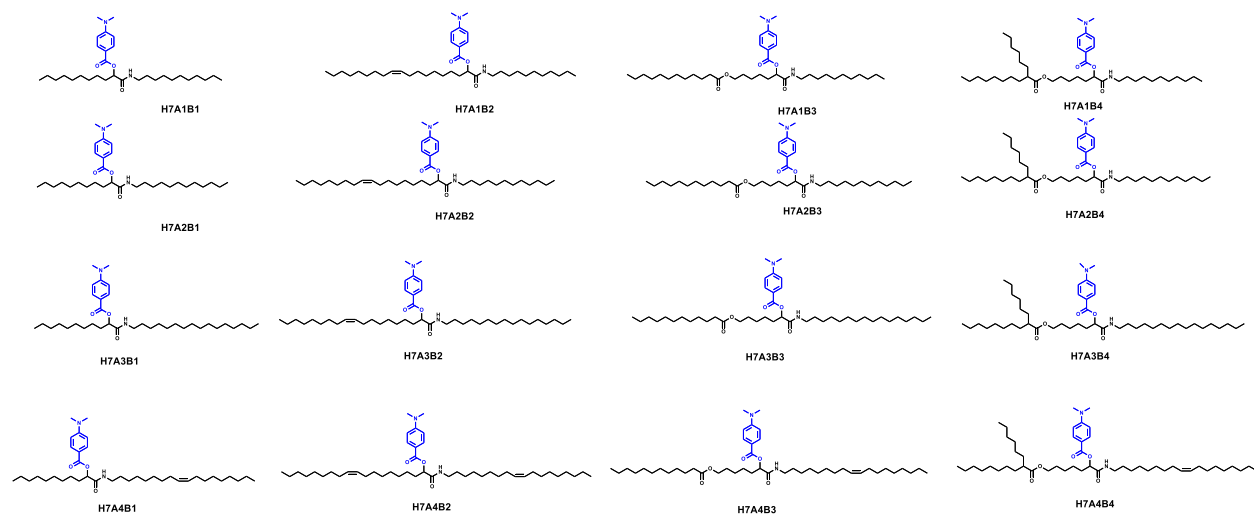


Figure S7. Lipid structures based on the head group 4-(dimethylamino) benzoic acid (H7) and synthesized by the Passerini reaction.

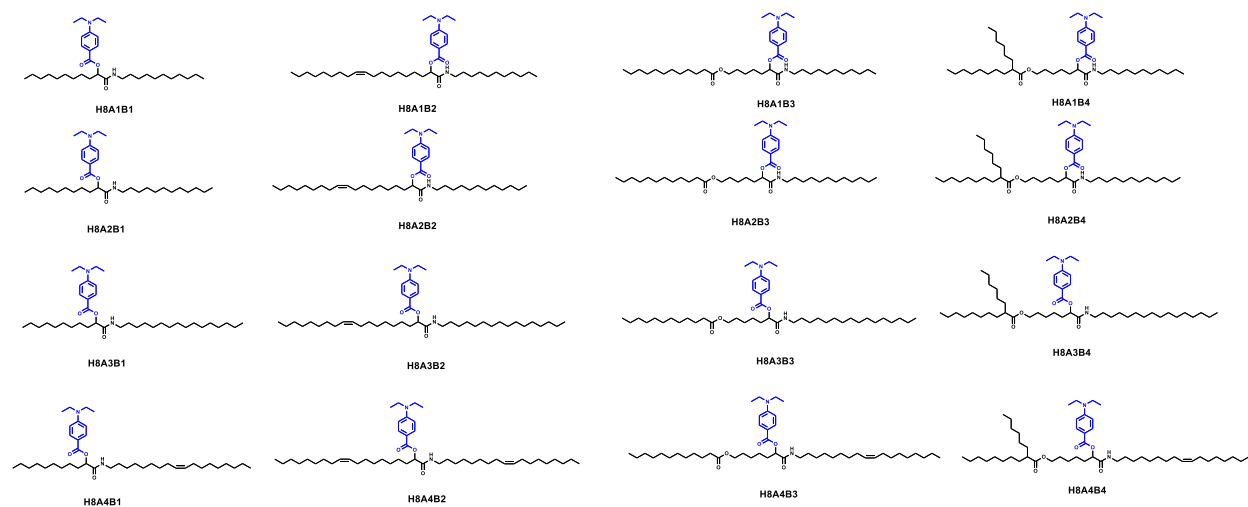


Figure S8. Lipid structures based on the head group 4-(diethylamino) benzoic acid (H8) and synthesized by the Passerini reaction.

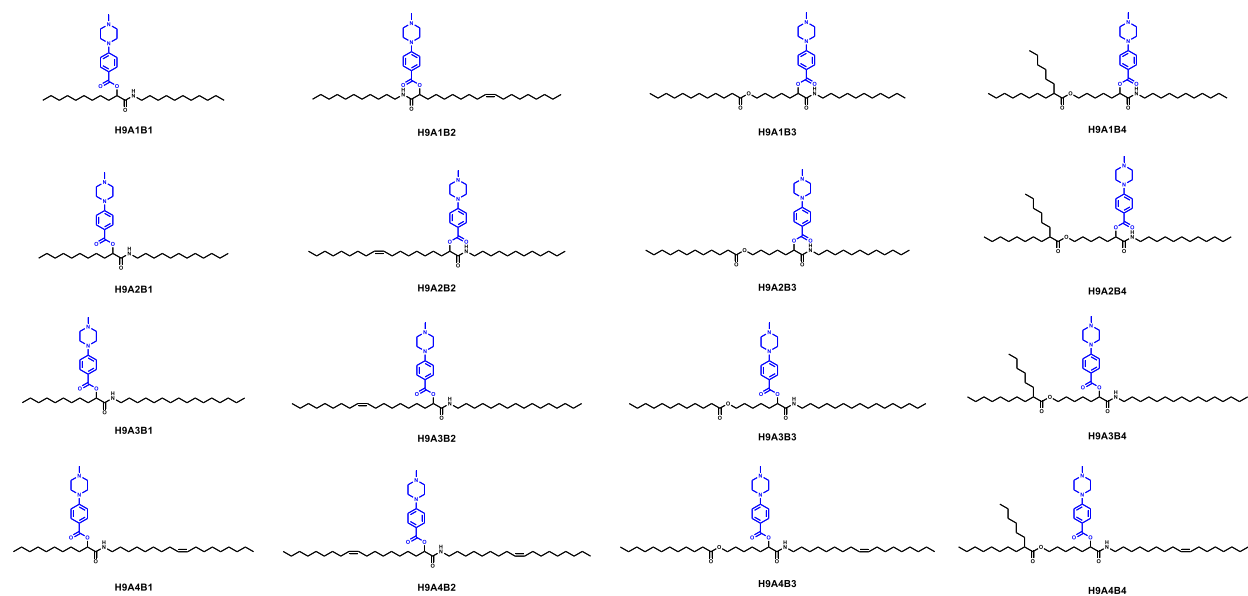


Figure S9. Lipid structures based on the head group 4-(4-methylpiperazin-1-yl) benzoic acid (H9) and synthesized by the Passerini reaction.

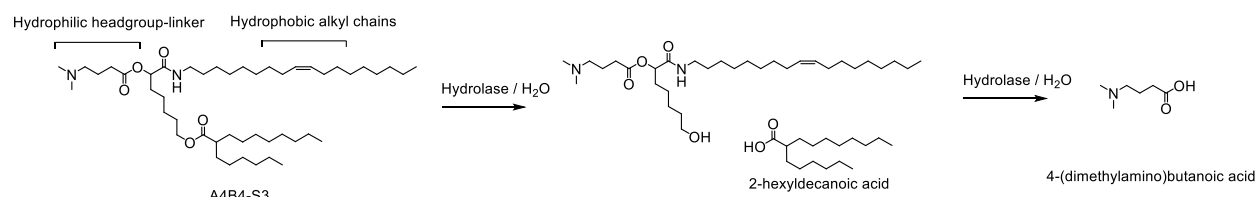


Figure S10. Proposed esterase-mediated degradation pathways for lipid A4B4-S3.

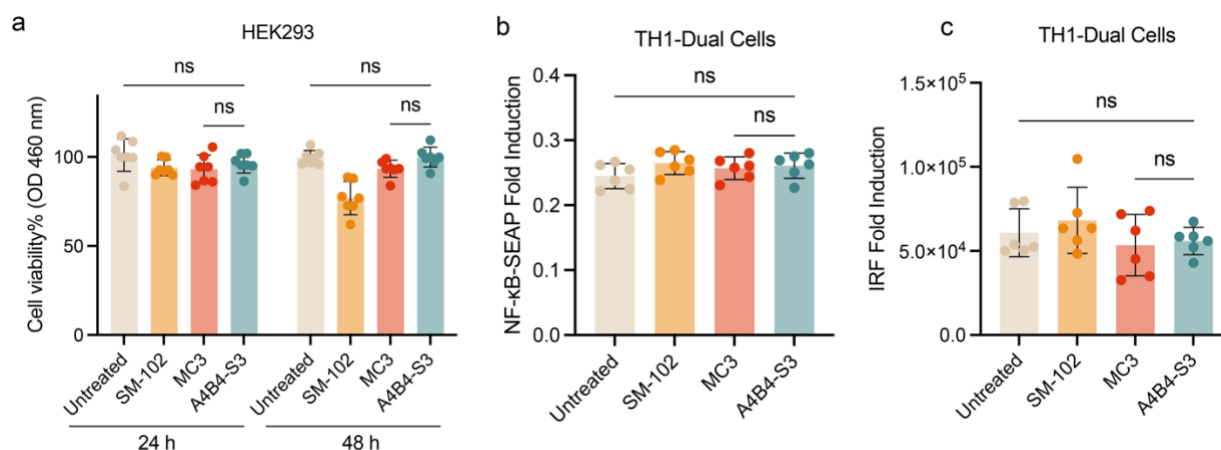


Figure S11. Cytotoxicity assay using WST-8 and immunogenicity evaluation through NF-κB-SEAP and IRF-Lucia luciferase reporter monocytic. WST-8 assay measures cell viability by detecting metabolic activity, while NF-κB-SEAP and IRF-Lucia assays assess immune activation via SEAP and luciferase expression, respectively.

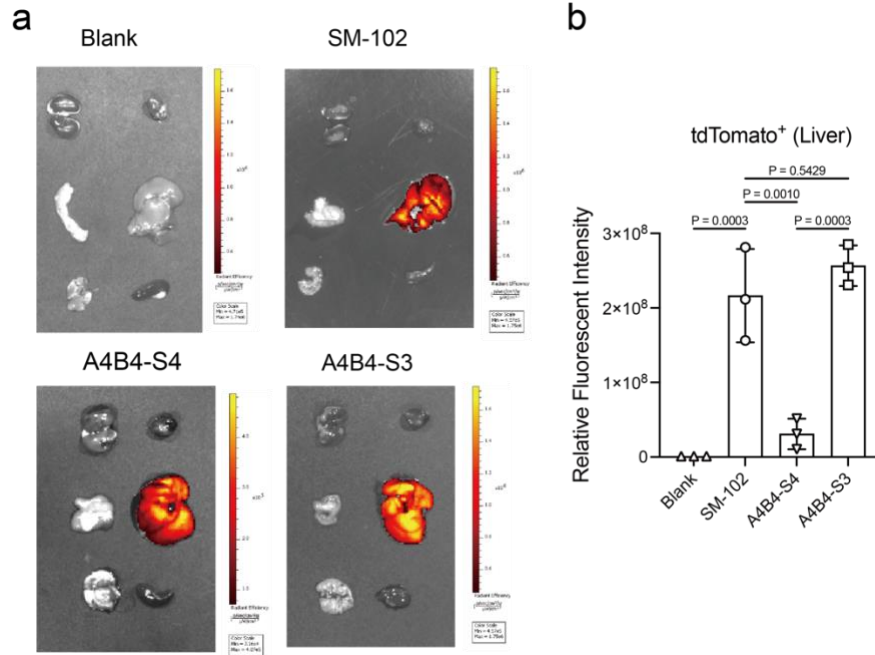


Figure S12. IVIS imaging for Ai9 cre reporter mice. (a) IVIS images of Cg-Gt(ROSA)26Sor^{tm9}(CAG-tdTomato)^{Hze/J} mice Cre reporter mice organs after cre-mRNA LNP by I.V. injection. (0.5 mg kg⁻¹) (b) Comparison of the relative fluorescence intensity of livers after I.V. injection of cre-mRNA LNP. (tdTomato: Ex:568 nm, Em:581 nm)

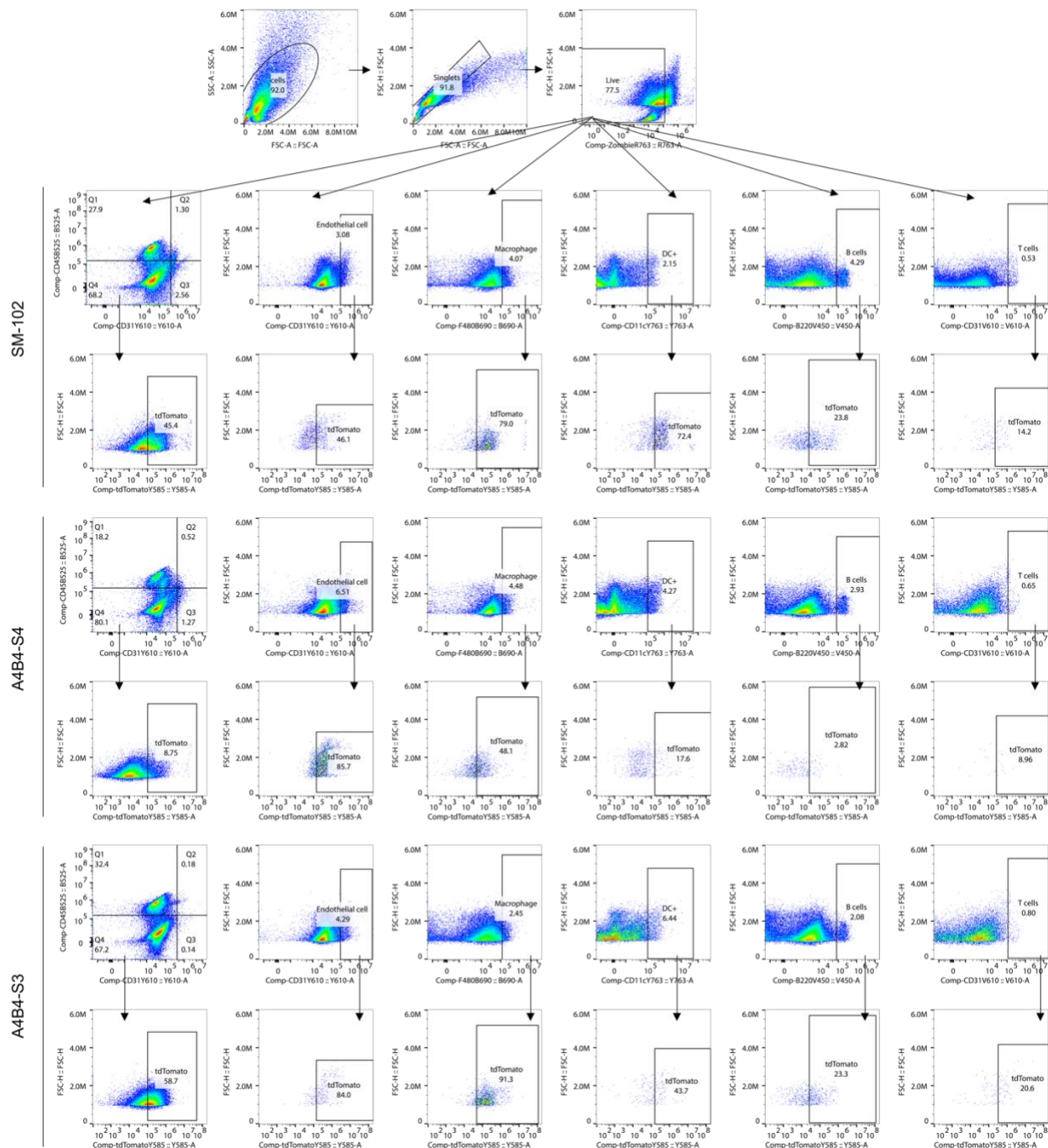


Figure S13. Flow cytometry gating strategy for liver hepatocytes, endothelial, macrophages, dendritic cells, B cells and T cells. Single cells prepared from Ai9 mouse livers. Liver cells were stained with antibodies against cell surface makers and detected by flow cytometry. Shown are the gating strategies of hepatocytes, endothelial, macrophages, dendritic cells, B cells and T cells in cell suspension (n=3).

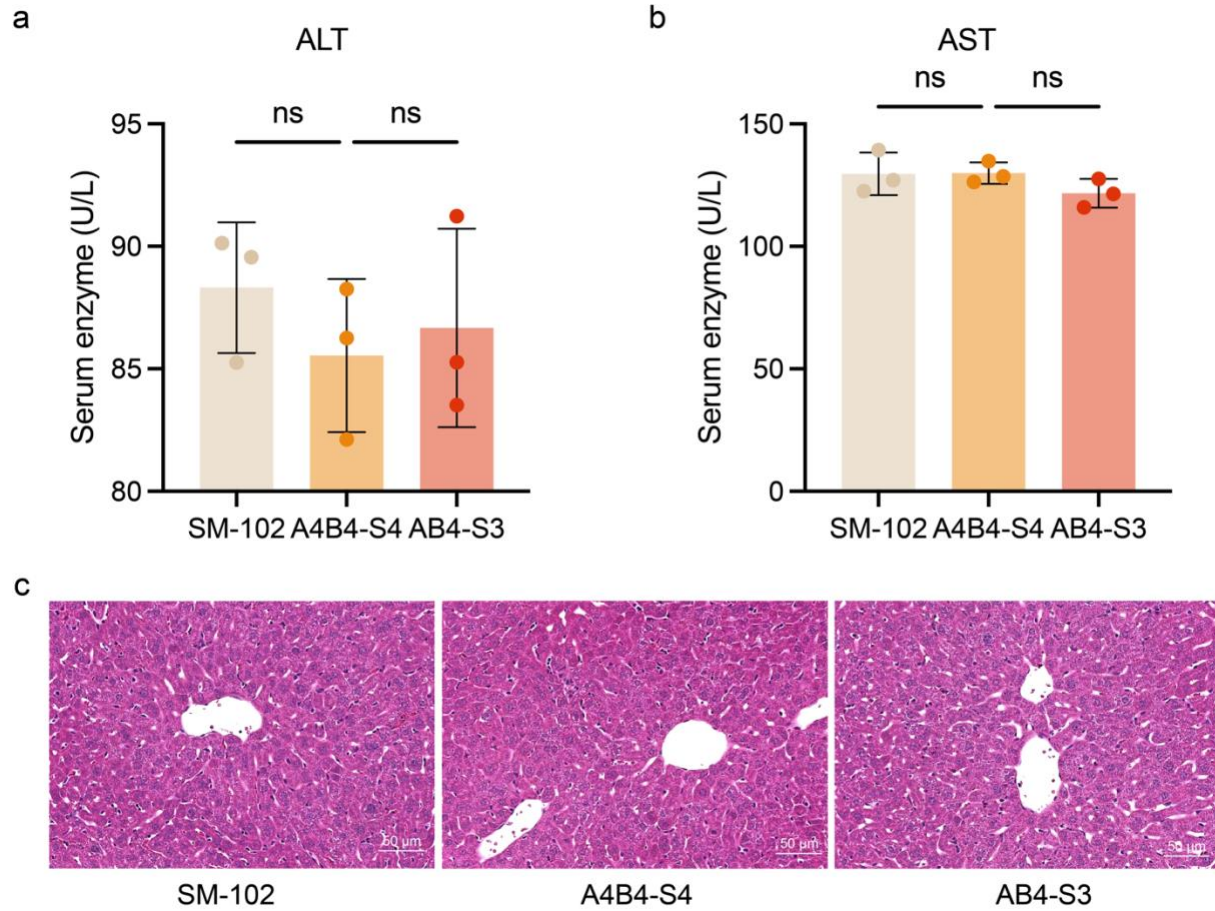


Figure S14. *In vivo* liver toxicity after repeated EPO mRNA LNP dose. (a, b) Serum enzyme levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST) by different groups (SM-102, A4B4-S4 and A4B4-S3 EPO mRNA LNP). (c) Hematoxylin and eosin (H&E) staining on liver sections after the repeated doses.

203 **Table S1. Physicochemical properties and characterization data for ionizable lipid and corresponding LNPs.**

Ionizable lipid			Lipid nanoparticles		
Name	Chemical Formula	Molecular Weight (MW)	Z-Ave (d.nm)	PDI	EE%
H1A1B1	C26H52N2O3	440.71	173.6	0.184	78.45%
H1A1B2	C34H66N2O3	550.91	182.6	0.209	88.62%
H1A1B3	C34H66N2O5	582.91	129.2	0.139	85.73%
H1A1B4	C38H74N2O5	639.02	125.4	0.155	85.71%
H1A2B1	C27H54N2O3	454.74	126.6	0.193	76.70%
H1A2B2	C35H68N2O3	564.94	130.1	0.176	76.85%
H1A2B3	C35H68N2O5	596.94	132.1	0.222	87.01%
H1A2B4	C39H76N2O5	653.05	130.7	0.200	75.02%
H1A3B1	C31H62N2O3	510.85	138.2	0.096	82.51%
H1A3B2	C39H76N2O3	621.05	134.3	0.164	82.63%
H1A3B3	C39H76N2O5	653.05	133.7	0.158	85.37%
H1A3B4	C43H84N2O5	709.15	135.0	0.156	84.41%
H1A4B1	C33H64N2O3	536.89	136.4	0.196	87.79%
H1A4B2	C41H78N2O3	647.09	135.3	0.030	91.06%
H1A4B3	C41H78N2O5	679.08	131.1	0.175	87.42%
H1A4B4	C45H86N2O5	735.19	129.5	0.168	77.58%
H2A1B1	C27H54N2O3	454.74	134.6	0.175	78.22%
H2A1B2	C35H68N2O3	564.94	135.9	0.075	79.32%
H2A1B3	C35H68N2O5	596.94	137.1	0.158	73.27%
H2A1B4	C39H76N2O5	653.05	138.3	0.204	94.82%
H2A2B1	C28H56N2O3	468.77	134.7	0.189	76.91%
H2A2B2	C36H70N2O3	578.97	129.4	0.157	88.42%
H2A2B3	C36H70N2O5	610.96	137.2	0.170	76.48%
H2A2B4	C40H78N2O5	667.07	137.7	0.133	78.25%
H2A3B1	C32H64N2O3	524.88	140.8	0.300	85.92%
H2A3B2	C40H78N2O3	635.07	137.9	0.092	78.59%
H2A3B3	C40H78N2O5	667.07	131.3	0.130	92.45%
H2A3B4	C44H86N2O5	723.18	137.5	0.159	95.59%
H2A4B1	C34H66N2O3	550.91	137.0	0.092	78.41%
H2A4B2	C42H80N2O3	661.11	128.0	0.215	94.65%
H2A4B3	C42H80N2O5	693.11	134.6	0.141	83.02%
H2A4B4	C46H88N2O5	749.22	139.4	0.154	86.24%
H3A1B1	C28H56N2O3	468.77	133.2	0.220	90.10%
H3A1B2	C36H70N2O3	578.97	141.1	0.080	84.81%
H3A1B3	C36H70N2O5	610.96	135.9	0.114	75.59%
H3A1B4	C40H78N2O5	667.07	132.7	0.205	94.86%
H3A2B1	C29H58N2O3	482.79	132.3	0.161	93.62%
H3A2B2	C37H72N2O3	592.99	135.1	0.214	94.48%
H3A2B3	C37H72N2O5	624.99	130.9	0.094	81.03%

H3A2B4	C41H80N2O5	681.10	122.6	0.932	92.78%
H3A3B1	C33H66N2O3	538.90	131.3	0.203	93.61%
H3A3B2	C41H80N2O3	649.10	128.1	0.191	77.82%
H3A3B3	C41H80N2O5	681.10	132.7	0.151	79.54%
H3A3B4	C45H88N2O5	737.21	131.8	0.222	84.51%
H3A4B1	C35H68N2O3	564.94	130.4	0.189	76.53%
H3A4B2	C43H82N2O3	675.14	134.3	0.187	88.40%
H3A4B3	C43H82N2O5	707.14	305.5	0.903	96.27%
H3A4B4	C47H90N2O5	763.25	130.6	0.212	91.87%
H4A1B1	C29H56N2O3	480.78	128.9	0.162	80.98%
H4A1B2	C37H70N2O3	590.98	134.7	0.187	93.00%
H4A1B3	C37H70N2O5	622.98	138.7	0.192	85.17%
H4A1B4	C41H78N2O5	679.08	134.6	0.181	79.15%
H4A2B1	C30H58N2O3	494.81	134.0	0.122	88.98%
H4A2B2	C38H72N2O3	605.00	140.2	0.200	85.06%
H4A2B3	C38H72N2O5	637.00	133.2	0.099	79.60%
H4A2B4	C42H80N2O5	693.11	129.4	0.175	87.66%
H4A3B1	C34H66N2O3	550.91	130.4	0.234	76.11%
H4A3B2	C42H80N2O3	661.11	136.7	0.077	91.98%
H4A3B3	C42H80N2O5	693.11	130.9	0.170	75.62%
H4A3B4	C46H88N2O5	749.22	129.8	0.139	89.22%
H4A4B1	C36H68N2O3	576.95	130.5	0.184	96.09%
H4A4B2	C44H82N2O3	687.15	237.0	0.715	75.67%
H4A4B3	C44H82N2O5	719.15	127.7	0.230	83.97%
H4A4B4	C48H90N2O5	775.26	128.0	0.181	86.73%
H5A1B1	C30H58N2O3	494.81	133.3	0.198	96.87%
H5A1B2	C38H72N2O3	605.00	124.6	0.203	78.05%
H5A1B3	C38H72N2O5	637.00	132.0	0.173	92.89%
H5A1B4	C42H80N2O5	693.11	132.4	0.134	88.38%
H5A2B1	C31H60N2O3	508.83	129.5	0.100	86.08%
H5A2B2	C39H74N2O3	619.03	131.1	0.195	83.26%
H5A2B3	C39H74N2O5	651.03	157.1	0.625	94.02%
H5A2B4	C43H82N2O5	707.14	129.8	0.123	80.01%
H5A3B1	C35H68N2O3	564.94	130.6	0.002	85.02%
H5A3B2	C43H82N2O3	675.14	131.3	0.020	87.23%
H5A3B3	C43H82N2O5	707.14	130.1	0.119	79.35%
H5A3B4	C47H90N2O5	763.25	126.4	0.102	92.55%
H5A4B1	C37H70N2O3	590.98	170.4	0.535	85.66%
H5A4B2	C45H84N2O3	701.18	134.5	0.101	84.55%

H5A4B3	C45H84N2O5	733.18	132.0	0.122	87.04%
H5A4B4	C49H92N2O5	789.28	149.7	0.124	93.68%
H6A1B1	C29H56N2O3	480.78	139.2	0.152	76.74%
H6A1B2	C37H70N2O3	590.98	134.8	0.111	86.40%
H6A1B3	C37H70N2O5	622.98	131.9	0.033	90.86%
H6A1B4	C41H78N2O5	679.08	131.9	0.136	92.84%
H6A2B1	C30H58N2O3	494.81	124.5	0.167	96.54%
H6A2B2	C38H72N2O3	605.00	129.1	0.089	79.07%
H6A2B3	C38H72N2O5	637.00	134.6	0.132	89.48%
H6A2B4	C46H88N2O5	693.11	161.4	0.532	94.89%
H6A3B1	C34H66N2O3	550.91	134.4	0.076	91.54%
H6A3B2	C42H80N2O3	661.11	129.7	0.114	94.02%
H6A3B3	C42H80N2O5	693.11	169.9	0.198	80.94%
H6A3B4	C46H88N2O5	749.22	128.1	0.220	85.52%
H6A4B1	C36H68N2O3	576.95	175.6	0.165	88.66%
H6A4B2	C44H82N2O3	687.15	134.8	0.121	83.37%
H6A4B3	C44H82N2O5	719.15	125.7	0.134	93.08%
H6A4B4	C48H90N2O5	775.26	137.6	0.143	87.11%
H7A1B1	C31H54N2O3	502.78	128.3	0.170	95.77%
H7A1B2	C39H68N2O3	612.98	133.1	0.111	83.99%
H7A1B3	C39H68N2O5	644.98	134.6	0.109	84.57%
H7A1B4	C43H76N2O5	701.09	129.3	0.207	88.59%
H7A2B1	C32H56N2O3	516.81	134.0	0.208	92.95%
H7A2B2	C36H64N2O3	572.92	129.4	0.164	76.70%
H7A2B3	C40H70N2O3	627.01	138.3	0.355	89.49%
H7A2B4	C44H78N2O5	715.12	124.2	0.181	89.96%
H7A3B1	C44H78N2O3	683.12	126.4	0.165	75.68%
H7A3B2	C44H78N2O5	715.12	128.8	0.188	92.49%
H7A3B3	C40H70N2O5	659.01	132.8	0.178	82.94%
H7A3B4	C48H86N2O5	771.23	134.4	0.194	90.32%
H7A4B1	C38H66N2O3	598.96	127.5	0.156	95.85%
H7A4B2	C46H80N2O3	709.16	130.4	0.231	78.99%
H7A4B3	C46H80N2O5	741.16	131.8	0.206	94.80%
H7A4B4	C50H88N2O5	797.26	134.6	0.173	83.80%
H8A1B1	C33H58N2O3	530.84	134.0	0.124	85.47%
H8A1B2	C41H72N2O3	641.04	126.8	0.148	95.54%
H8A1B3	C41H72N2O5	673.04	132.8	0.204	91.46%
H8A1B4	C45H80N2O5	729.14	126.6	0.119	92.69%
H8A2B1	C34H60N2O3	544.87	127.8	0.192	94.41%

H8A2B2	C42H74N2O3	655.06	127.4	0.155	86.61%
H8A2B3	C42H74N2O5	687.06	131.5	0.072	76.58%
H8A2B4	C46H82N2O5	743.17	133.7	0.156	76.64%
H8A3B1	C38H68N2O3	600.97	128.8	0.075	80.25%
H8A3B2	C46H82N2O3	711.17	128.0	0.149	78.53%
H8A3B3	C46H82N2O5	743.17	132.8	0.166	75.61%
H8A3B4	C50H90N2O5	799.28	127.8	0.153	80.89%
H8A4B1	C40H70N2O3	627.01	129.3	0.094	85.81%
H8A4B2	C48H84N2O3	737.21	126.7	0.165	80.54%
H8A4B3	C48H84N2O5	769.21	129.7	0.105	92.71%
H8A4B4	C52H92N2O5	825.32	130.5	0.165	95.51%
H9A1B1	C34H59N3O3	557.86	128.5	0.213	94.22%
H9A1B2	C42H73N3O3	668.06	132.2	0.083	82.97%
H9A1B3	C42H73N3O5	700.06	132.2	0.106	79.99%
H9A1B4	C46H81N3O5	756.17	127.4	0.203	75.69%
H9A2B1	C35H61N3O3	571.89	130.9	0.085	80.40%
H9A2B2	C43H75N3O3	682.09	127.9	0.154	77.55%
H9A2B3	C43H75N3O5	714.09	129.9	0.174	81.54%
H9A2B4	C47H83N3O5	770.20	131.7	0.143	76.48%
H9A3B1	C39H69N3O3	628.00	131.2	0.082	86.47%
H9A3B2	C47H83N3O3	738.20	134.3	0.078	89.40%
H9A3B3	C47H83N3O5	770.20	129.8	0.209	84.95%
H9A3B4	C51H91N3O5	826.30	132.2	0.170	86.68%
H9A4B1	C41H71N3O3	654.04	128.4	0.215	86.66%
H9A4B2	C49H85N3O3	764.24	126.0	0.220	82.51%
H9A4B3	C49H85N3O5	796.24	132.0	0.182	90.68%
H9A4B4	C53H93N3O5	852.34	126.8	0.190	96.20%

205 **Table S2. Energy of combination and ionization potential calculation for ionizable lipids with Ribose-phosphate**
 206 **helix.**

Energy of combination and ionization potential									
$E_{comb.} = E_{AB} - E_A - E_B$	$A + B \rightarrow AB$								
opt: B3LYP-D3(BJ)/def2-SVP		A4B4-S3	A4B4-S4	Adenine (A)	Guanine (G)	Cytosine (C)	Uracil (U)		
M06-2X-D3(0)/ma-def2-TZVP	ϵ_{ele} / au	-2330.69	-2370.00	-1530.18	-1605.43	-1457.79	-1477.69		
opt: B3LYP-D3(BJ)/def2-SVP		A4B4-S3+A	A4B4-S3+G	A4B4-S3+C	A4B4-S3+U	A4B4-S4+A	A4B4-S4+G	A4B4-S4+C	A4B4-S4+U
M06-2X-D3(0)/ma-def2-TZVP	ϵ_{ele} / au	-3860.95	-3936.20	-3788.57	-3808.45	-3900.26	-3975.51	-3827.88	-3847.76
Binding Energy (T = 298.15 K, P = 1 atm)	$E_{combine} / kcal/mol$	-47.9	-50.28	-55.88	-42.56	-49.43	-52	-56.01	-44.73
	$E_{combine} / kJ/mol$	-200.43	-210.36	-233.82	-178.06	-206.82	-217.57	-234.36	-187.14
	$E_{combine} / eV$	-2.08	-2.18	-2.42	-1.85	-2.14	-2.25	-2.43	-1.94
IP = $E_{A^+} - E_A$	$A - e^- \rightarrow A^+$	A4B4-S3	A4B4-S3*	A4B4-S4	A4B4-S4*				
M06-2X-D3(0)/def2-TZVP	ϵ_{ele} / au	-2330.69	-2330.39	-2370.00	-2369.70				
Ionization Potential	$E_{IP} / kcal/mol$	188.93	188.40						
	$E_{IP} / kJ/mol$	790.47	788.25						
	E_{IP} / eV	8.19	8.17						

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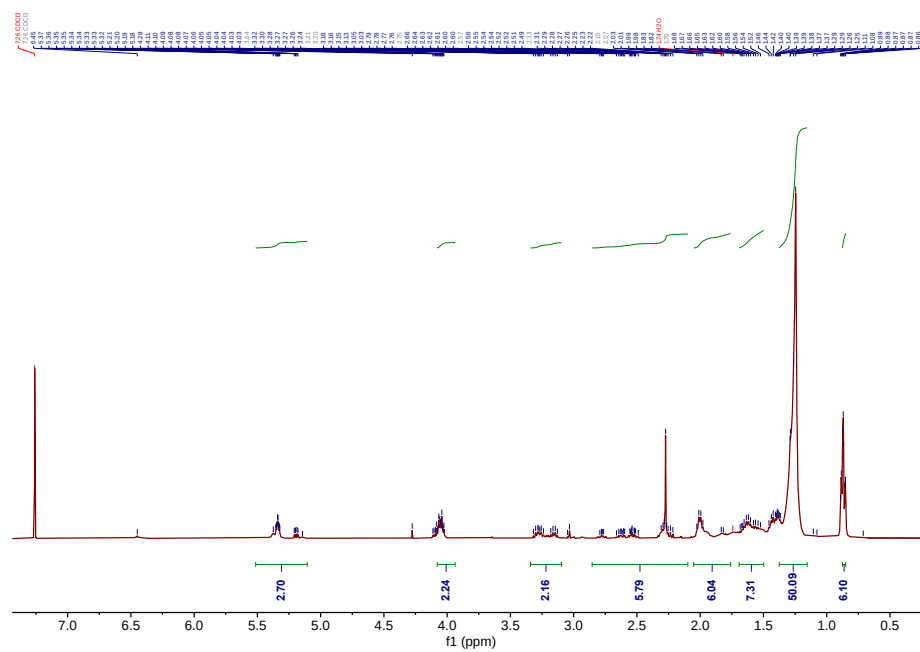
208 **Table S3. Antibodies for flow cytometry.**

Specificity	Fluorochrome	Supplier	Catalog
CD45	FITC	BioLegend	157214
CD31	Alexa Fluor® 594	BioLegend	102520
F4/80	PerCP/Cyanine5.5	BioLegend	123128
CD11c	PE/Cyanine7	BioLegend	117318
CD3	Brilliant Violet 605™	BioLegend	100237
CD45R/B220	Brilliant Violet 421™	BioLegend	103240

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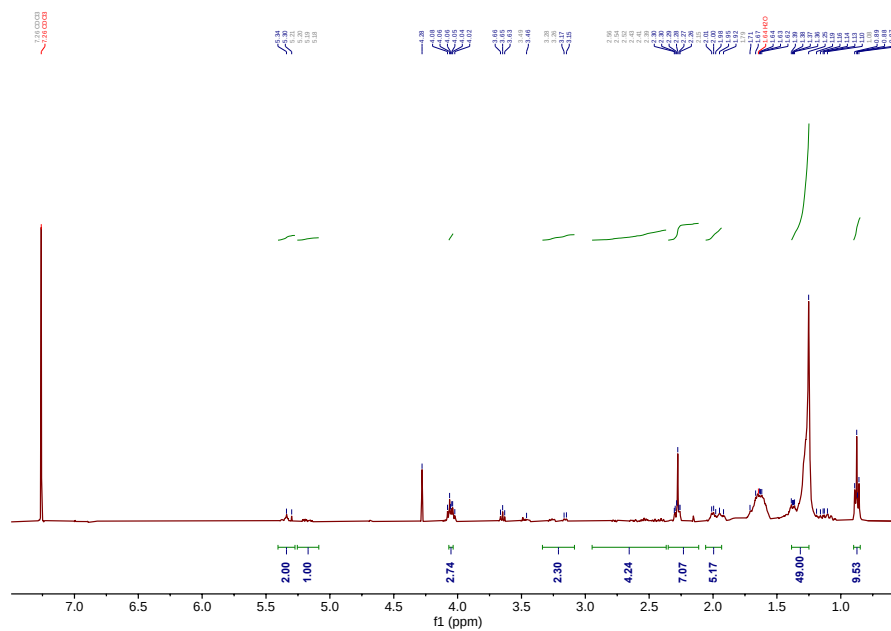
210 *Section C. Characterization Details of Synthetic Compounds*

211 ^1H NMR spectra of H2A4B3



212

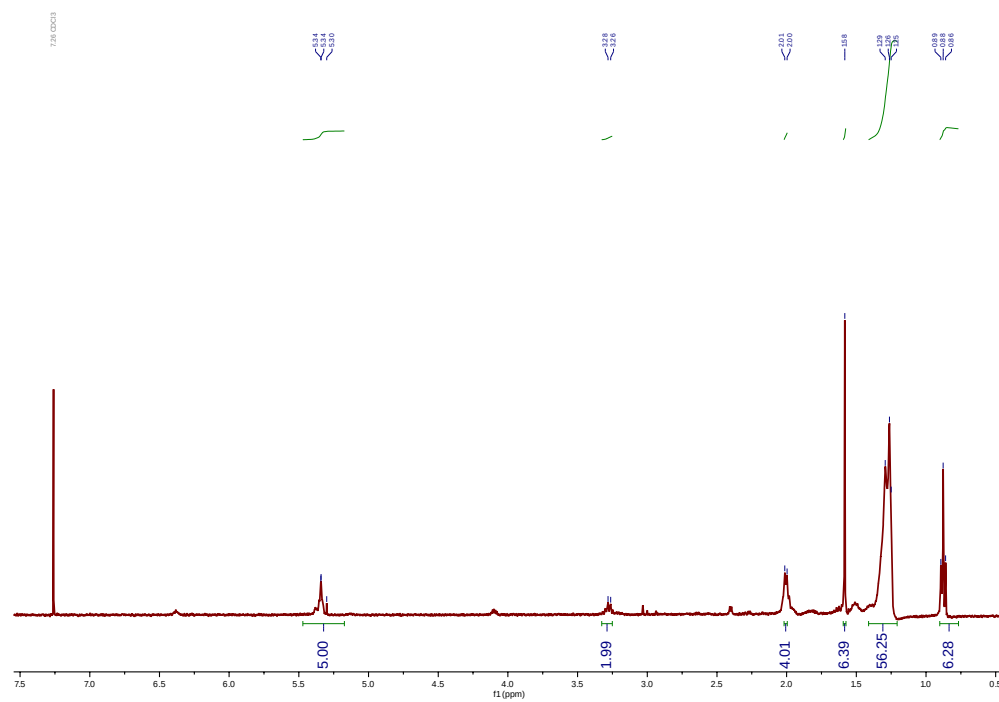
213 ^1H NMR spectra of H2A4B4



214

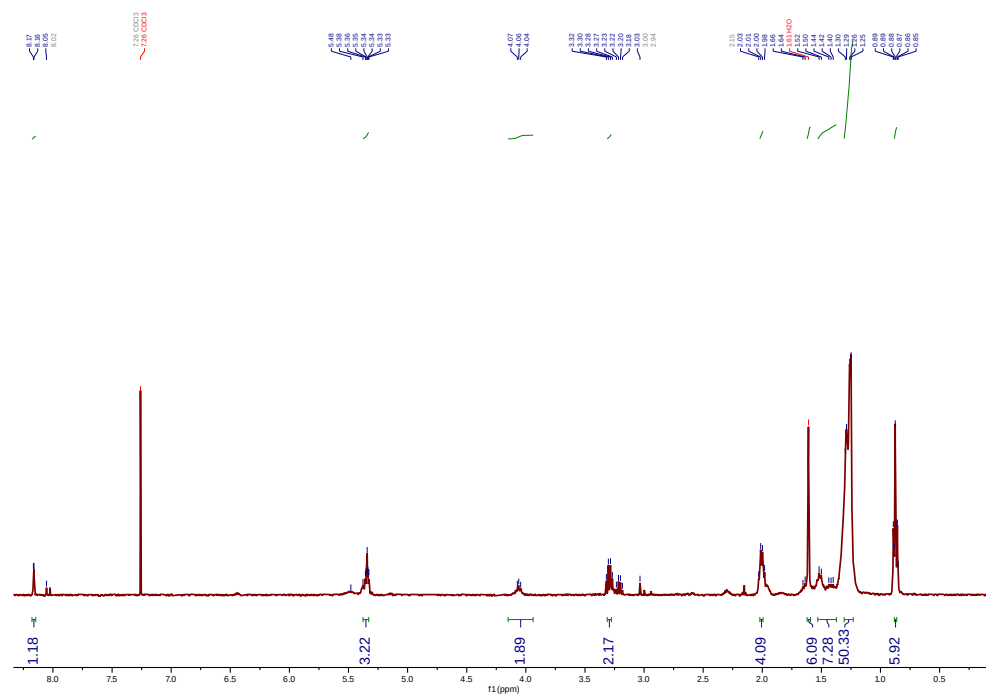
215

216 ^1H NMR spectra of H2A4-Ole



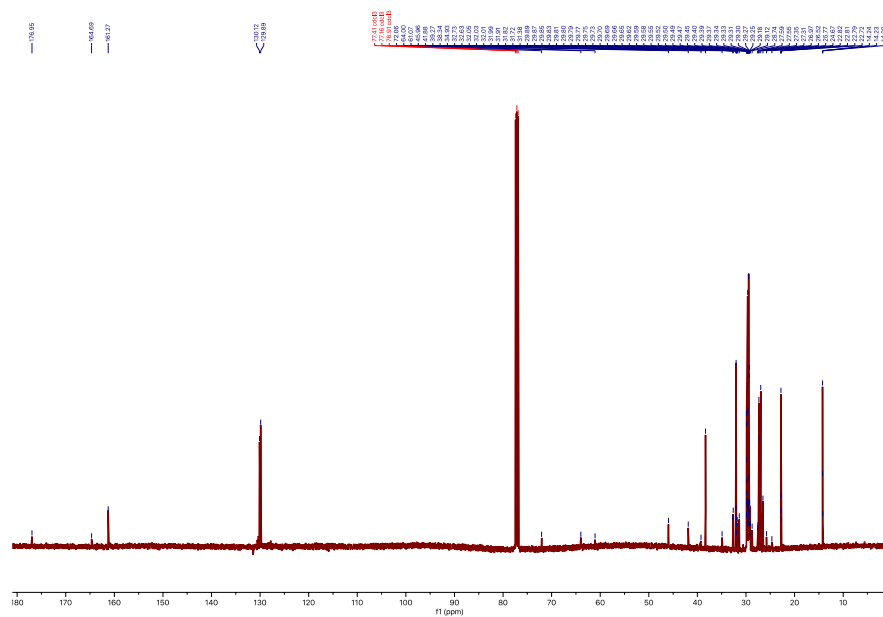
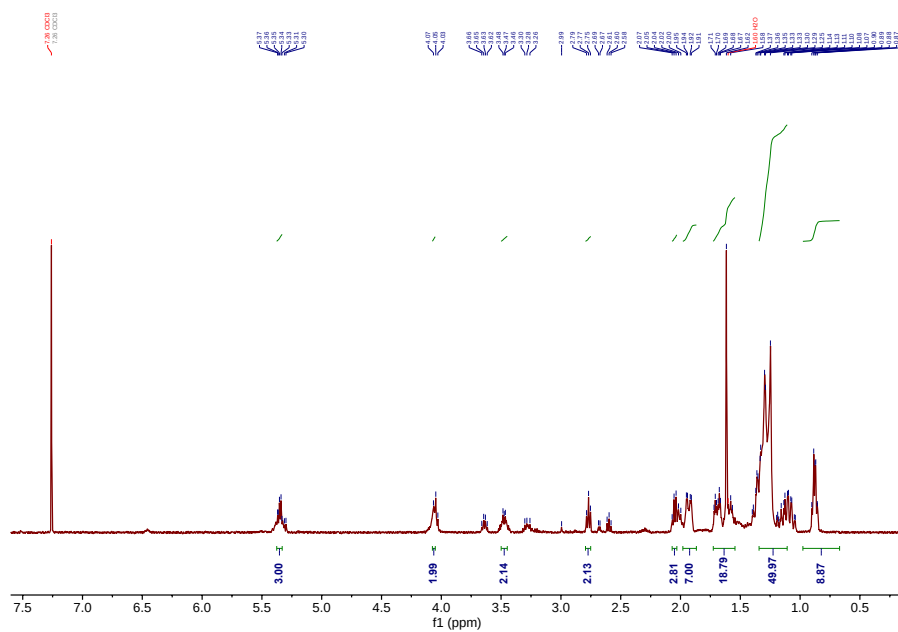
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218 ^1H NMR spectra of H2A4-Butyl



219

221

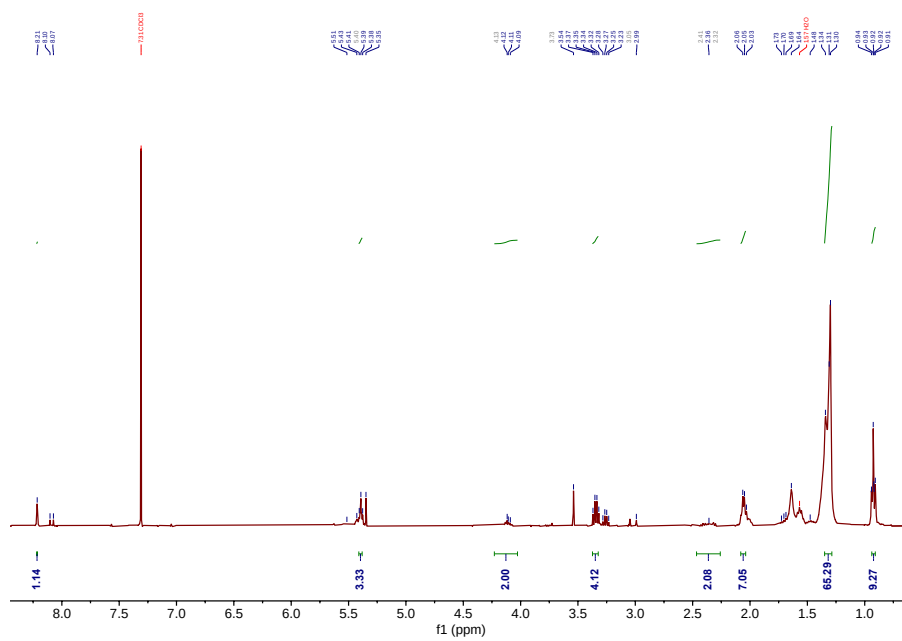


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224

225 ^1H NMR spectra of A4B3-S4



226

Section D. Reference

1. Chen, J. et al. Combinatorial design of ionizable lipid nanoparticles for muscle-selective mRNA delivery with minimized off-target effects. *Proceedings of the National Academy of Sciences* **120**, e2309472120 (2023).
2. Xu, Y. et al. AGILE platform: a deep learning powered approach to accelerate LNP development for mRNA delivery. *Nature Communications* **15**, 6305 (2024).